

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12384519
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Yamamoto; Kazuomi et al.

Leading-edge high-lift device, wing, and aircraft

Abstract

A leading-edge high-lift device according to an embodiment of the present invention is a leading-edge high-lift device that is deployable and retractable from/into a fixed leading edge of a main wing of an aircraft and includes: a leading edge portion; a trailing edge portion that forms a gap between a trailing edge portion and the main wing during deployment; a cusp portion formed at a lower edge of the leading edge portion; a lower surface portion formed between the cusp portion and the trailing edge portion; and a curved hump portion that is locally provided on a surface of the lower surface portion and protrudes toward the main wing in a cross-sectional shape perpendicular to a wing span direction of the main wing.

Inventors:	Yamamoto; Kazuomi (Tokyo, JP), Murayama; Mitsuhiro (Tokyo, JP), Yokokawa; Yuzuru (Tokyo, JP), Kohzai; Masataka (Tokyo, JP), Isotani; Kazuhide (Kobe, JP), Ueno; Yosuke (Kobe, JP), Hayama; Kenji (Kobe, JP), Hayashi; Kensuke (Nagoya, JP)
Applicant:	JAPAN AEROSPACE EXPLORATION AGENCY (Tokyo, JP); KAWASAKI JUKOGYO KABUSHIKI KAISHA (Kobe, JP); MITSUBISHI HEAVY INDUSTRIES, LTD. (Tokyo, JP)
Family ID:	84028235
Assignee:	JAPAN AEROSPACE EXPLORATION AGENCY (Tokyo, JP); KAWASAKI JUKOGYO KABUSHIKI KAISHA (Kobe, JP); MITSUBISHI HEAVY INDUSTRIES, LTD. (Tokyo, JP)
Appl. No.:	18/560854
Filed (or PCT Filed):	March 31, 2022
PCT No.:	PCT/JP2022/016742
PCT Pub. No.:	WO2022/239574
PCT Pub. Date:	November 17, 2022

Prior Publication Data

Document Identifier

US 20240246657 A1

Publication Date

Jul. 25, 2024

Foreign Application Priority Data

JP

2021-082182

May. 14, 2021

Publication Classification

Int. Cl.: B64C9/24 (20060101)

U.S. Cl.:

CPC B64C9/24 (20130101);

Field of Classification Search

CPC: B64C (9/24)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6457680	12/2001	Dobrzynski et al.	N/A	N/A
8424810	12/2012	Shmilovich et al.	N/A	N/A
9242720	12/2015	Turner et al.	N/A	N/A
2010/0084508	12/2009	Hirai et al.	N/A	N/A
2012/0097791	12/2011	Turner et al.	N/A	N/A
2012/0286101	12/2011	Hirai et al.	N/A	N/A
2019/0112029	12/2018	Turner	N/A	B64C 3/50

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2009-6987	12/2008	JP	N/A
4699487	12/2010	JP	N/A
2011-162154	12/2010	JP	N/A

OTHER PUBLICATIONS

Pott-Pollenske, M. et al., "Aerodynamic and Acoustic Design of Silent Leading Edge Devices," American Institute of Aeronautics and Astronautics, Jun. 16-20, 2014, pp. 1-15. cited by applicant
International Search Report dated May 24, 2022 in International Application No. PCT/JP2022/016742. cited by applicant

Primary Examiner: Xavier; Valentina

Attorney, Agent or Firm: Saliwanchik, Lloyd & Eisenschenk

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. national stage application of International Patent Application No. PCT/JP2022/016742, filed Mar. 31, 2022, which claims the benefit under 35 U.S.C. § 119 of Japanese Application No. 2021-082182, filed May 14, 2021, the disclosures of each of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The present invention relates to a leading-edge high-lift device mounted on an aircraft's wing and to a wing and an aircraft each including the same.

BACKGROUND ART

(3) High-lift devices are deployed from an aircraft wing in order to perform low-speed flight when an aircraft takes off or lands at an airport. Leading-edge high-lift devices, e.g., slats attached to a fixed leading edge that is a leading edge of the main wing of the aircraft, especially in an airliner, for example, generate high lift during low-speed flight.

(4) The slats increase the upper limit (maximum lift) of lift of the aircraft wing with gaps formed between the slats and the fixed leading edge. However, they also generate high aerodynamic noise under flight conditions in an approach and landing phase. The noise comes from turbulence in reverse flow regions formed at concaves (coves) formed on lower surfaces of the slats due to constraints for retracting the slats in the aircraft wing. Reducing pressure fluctuations caused by such turbulence allows slat noise reduction, but it is necessary to achieve both requirements for flight performance, such as maximum lift on the slats and structures and deployment mechanisms.

(5) The concept “cove filler” that adds a curved shape along a shear layer in the reverse flow region to a slat lower surface for avoiding the reverse flow region in the cove is known as a typical technology for slat noise reduction (see Patent Literatures 1 to 3).

(6) Other known methods are, for example, a method (see Patent Literature 4) of providing a cusp of a slat lower surface where the reverse flow region is generated with a “serration” that actively mixes a shear layer in the cove for preventing large pressure fluctuations, a method (see Patent Literature 5) of providing a slat lower surface with an “inclined plate” for changing an angle of the shear layer in the reverse flow region that impinges on the slat lower surface, and a technology (see Non-Patent Literature 1) of designing a very long chord slat, which is called VLCS.

CITATION LIST

Patent Literature

(7) Patent Literature 1: U.S. Pat. No. 6,457,680 Patent Literature 2: U.S. Pat. No. 9,242,720 Patent Literature 3: U.S. Pat. No. 8,424,810 Patent Literature 4: Japanese Patent Application Laid-open No. 2011-162154 Patent Literature 5: Japanese Patent No. 4699487

Non-Patent Literature

(8) Non-Patent Literature 1: Pott-Polenske, M., Wild, J., and Bertsch, L., “Aerodynamic and Acoustic Design of Silent Leading Edge Devices,” AIAA Paper 2014-2076, 20th AIAA/CEAS Aeroacoustics Conference, 16-20 Jun. 2014.

DISCLOSURE OF INVENTION

Technical Problem

(9) The technologies described in Patent Literatures 1 to 3 achieve significant noise reduction. However, the need for large cove shape deformation during retraction in the aircraft wing introduces complex mechanisms and increased aircraft weight.

(10) Moreover, the technology disclosed in Patent Literature 4 reduces low-frequency peak sound but it increases high-frequency noise; thus, the technology cannot sufficiently reduce noise.

(11) In addition, the technology disclosed in Patent Literature 5 requires less shape deformation

than with the cove filler but cannot sufficiently reduce noise.

(12) The technology disclosed in Non-Patent Literature 1 requires the slat length to be nearly double the conventional slat, so the technology cannot be achieved under constraints on the conventional wing structure.

(13) In view of the circumstances mentioned above, it is an objective of the present invention to provide a leading-edge high-lift device that can achieve both noise reduction and requirements for flight performance and a wing and an aircraft each including the same.

Solution to Problem

(14) A leading-edge high-lift device according to an embodiment of the present invention is a leading-edge high-lift device that is deployable and retractable from/into a fixed leading edge of a main wing of an aircraft and includes: a leading edge portion; a trailing edge portion that forms a gap between a trailing edge portion and the main wing during deployment; a cusp portion formed at a lower edge of the leading edge portion; a lower surface portion formed between the cusp portion and the trailing edge portion; and a curved hump portion that is locally provided on a surface of the lower surface portion and protrudes toward the main wing in a cross-sectional shape perpendicular to a wing span direction of the main wing.

(15) Since the leading-edge high-lift device includes the hump portion locally provided on the surface of the lower surface portion, the distance between a reattachment point of a turbulent shear layer on the lower surface portion and the trailing edge portion can be set to be longer than that in a case where no hump portion is provided. Accordingly, pressure fluctuations at the trailing edge portion can be attenuated, and noise reduction can be achieved.

(16) Assuming that, in a cross-sectional shape perpendicular to the wing span direction of the main wing, a distance between the leading edge portion and the fixed leading edge, which is measured on the chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is a first distance and a distance between a top portion of the hump portion and the fixed leading edge, which is measured on the chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is a second distance, the second distance may be 10% or more and 40% or less of the first distance.

(17) Assuming that in a cross-sectional shape perpendicular to the wing span direction of the main wing, a distance between the leading edge portion and the fixed leading edge, which is measured on the chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is a first distance, a maximum thickness of the hump portion from the surface of the lower surface portion may be 5% or more and 15% or less of the first distance.

(18) Assuming that in a cross-sectional shape perpendicular to the wing span direction of the main wing, a distance between the leading edge portion and the fixed leading edge, which is measured on the chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is a first distance, an average curvature of a top portion of the hump portion may be double or more than double and ten times or less than ten times a multiplicative inverse of the first distance.

(19) Assuming that in a cross-sectional shape perpendicular to the wing span direction of the main wing, a distance between the leading edge portion and the fixed leading edge, which is measured on the chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is a first distance, an amount of overlap that is a distance between the trailing edge portion and the fixed leading edge, which is measured on the chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, may be 35% or less of the first distance.

(20) The leading-edge high-lift device may further include an adjustment member that is attached to at least a part of the trailing edge portion and adjusts the amount of overlap.

(21) The hump portion may be configured to expand during deployment from the fixed leading edge and be deformable along the fixed leading edge during retraction into the fixed leading edge.

(22) A leading-edge high-lift device according to another embodiment of the present invention is a leading-edge high-lift device that is deployable and retractable from/into a fixed leading edge of a main wing of an aircraft and includes: a leading edge portion; a trailing edge portion that forms a gap between a trailing edge portion and the main wing during deployment; a cusp portion formed at a lower edge of the leading edge portion; a lower surface portion formed between the cusp portion and the trailing edge portion; and a trailing edge extension portion provided on at least a part of the trailing edge portion.

(23) Assuming that, in a cross-sectional shape perpendicular to the wing span direction of the main wing, a distance between the leading edge portion and the fixed leading edge, which is measured on the chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is a first distance, an amount of overlap that is a distance between a tip end portion of the trailing edge extension portion and the fixed leading edge, which is measured on the chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is 10% or more and 35% or less of the first distance, and the first distance is 10% or more and 20% or less of the chord length of an aircraft wing during retraction of the leading-edge high-lift device into the fixed leading edge.

(24) This leading-edge high-lift device can increase the distance between the reattachment point of the turbulent shear layer on the lower surface portion and the trailing edge portion. Accordingly, pressure fluctuations at the trailing edge portion can be attenuated, and noise reduction can be achieved.

(25) The trailing edge extension portion may be an adjustment member that is attached to the trailing edge portion and adjusts the amount of overlap.

Advantageous Effects of Invention

(26) In accordance with the present invention, both noise reduction and flight performance requirements can be achieved.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 A partial perspective view of a configuration example of an aircraft wing from an upper surface side.

(2) FIG. 2 A partial perspective view of the aircraft wing as viewed from a lower surface side.

(3) FIG. 3 A partial perspective view of leading-edge high-lift devices as viewed from the lower surface side.

(4) FIG. 4 A cross-sectional view of a reference-shape slat (reference slat), which is perpendicular to the wing span direction of a main wing.

(5) FIG. 5 A numerical simulation result showing an example of a flow field in the reference slat shown in FIG. 4 in a landing phase.

(6) FIG. 6 A cross-sectional view of the slat, which is perpendicular to the wing span direction according to Configuration Example 1 of the present embodiment.

(7) FIG. 7 A numerical simulation result showing an example of a flow field in the slat shown in FIG. 6 in a landing phase.

(8) FIG. 8 A cross-sectional view showing a retracted state of the slat shown in FIG. 6 in the main wing.

(9) FIG. 9 An explanatory view of the details of a hump portion of the slat shown in FIG. 6.

(10) FIG. 10 A cross-sectional view of a slat according to Configuration Example 2 of the present embodiment, which is perpendicular to the wing span direction.

(11) FIG. 11 A numerical simulation result showing an example of a flow field of the slat shown in FIG. 10 in a landing phase.

(12) FIG. 12 A similar cross-sectional view showing a modified example of a configuration of the slat shown in FIG. 10.

(13) FIG. 13 A numerical simulation result showing pressure fluctuations of a slat trailing edge portion, which cause noise generation, in which “A” shows an example of the reference slat, “B” shows an example of the slat according to Configuration Example 1, and “C” shows an example of the slat according to Configuration Example 2.

(14) FIG. 14 A numerical simulation result of overall sound pressure level (OASPL) directivity when slat noise is assessed.

(15) FIG. 15 A numerical simulation result of lift characteristics.

(16) FIG. 16 A graph describing shape effects associated with noise reduction of the slats according to Configuration Examples 1 and 2 compared to the overall sound pressure level of the reference slat shown in FIG. 14 in a ground direction (at 270 degrees).

(17) FIG. 17 A cross-sectional view of a Krueger flap as the leading-edge high-lift device according to the present embodiment, which is perpendicular to the wing span direction.

MODE(S) FOR CARRYING OUT THE INVENTION

(18) Hereinafter, an embodiment of the present invention will be described with reference to the drawings.

(19) [Overview of High-Lift Device]

(20) FIG. 1 is a partial perspective view of a configuration example of an aircraft wing (left wing) 100 as viewed from an upper surface side. FIG. 2 is a partial perspective view of the aircraft wing 100 as viewed from a lower surface side. FIG. 3 is a partial perspective view as viewed from the lower surface side of slats 20 constituting the aircraft wing 100.

(21) The aircraft wing 100 includes a main wing 10, the slats 20 disposed on the side of a leading edge 10a of the main wing 10, and flaps 30 disposed on the side of a trailing edge 10b of the main wing 10.

(22) It should be noted that the other aircraft wing (right wing) is configured to be similar to the aircraft wing 100.

(23) The slats 20 are configured to be deployable and retractable from/into the leading edge 10a of the main wing 10. The slats 20 are retracted into the leading edge 10a of the main wing 10 as shown in the figure in a cruise phase and deployed from the leading edge 10a of the main wing 10 by slat supporting devices 51 in a landing or take-off phase. The leading edge 10a of the main wing 10 refers to a region facing the slats 20 in the chord direction of the slats 20. It should be noted that in the following description, the leading edge 10a will be also referred to as a fixed leading edge 10a.

(24) The flaps 30 are configured to be deployable and retractable from/into the trailing edge 10b of the main wing 10. The flaps 30 are retracted into the trailing edge 10b of the main wing 10 as shown in the figure in a cruise phase and deployed from the trailing edge 10b of the main wing 10 through flap supporting devices 52 in a landing or take-off phase.

(25) The slats 20 are typically divided into a plurality of slats along the fixed leading edge 10a with the engine 40 provided therebetween. The length of each slat 20 in a wing span direction is arbitrarily set to a required length depending on the installed region. Also, the flaps 30 are typically divided into a plurality of flaps along the trailing edge 10b of the main wing 10, each having an arbitrary length. The slats 20 and the flaps 30 are made of, for example, a metal material such as an aluminum alloy or stainless steel or a composite material such as carbon fiber reinforced plastics (CFRP) or glass fiber reinforced plastics (GFRP).

(26) The slat 20 is one of the leading-edge high-lift devices. Providing a gap between the slat 20 and the main wing 10, which allows airflow to pass therethrough during deployment, increases maximum lift (upper limit of lift) of the aircraft wing 100 and increases a stall angle of attack of the aircraft wing 100. The size of the gap between the slat 20 and the main wing 10 is adjusted depending on a degree (angle) of deployment of the slat 20 from the main wing 10. Typically, the

slat **20** is more largely deployed in a landing phase than in a take-off phase. However, the gap between the slat **20** and the main wing **10** causes noise generation due to pressure fluctuations associated with turbulence in a reverse flow region.

(27) In view of this, in the present embodiment, the slat **20** is configured to suppress the noise generation due to pressure fluctuations caused by the gap between the slat **20** and the main wing **10** during deployment. Although it will be described later in detail, a slat having a baseline shape (hereinafter, also referred to as a reference slat **120**) that is a base for designing the slat **20** according to the present embodiment will be first described.

(28) It should be noted that the reference slat **120** corresponds to a slat with the conventional structure optimized only in terms of the aerodynamic performance.

(29) [Reference Slat]

(30) FIG. **4** is a cross-sectional view of the reference slat **120**, which is perpendicular to the wing span direction of the main wing **10**. Here, a state of the reference slat **120** in a landing phase when deployed at the maximum (fully deployed) from the fixed leading edge **10a** is shown.

(31) The reference slat **120** includes a leading edge portion **121**, a trailing edge portion **122**, a cusp portion **123**, a lower surface portion **124**, and an upper surface portion **126**. As shown in FIG. **4**, the cross-sectional shape of the reference slat **120** is a shape that forms a closed space surrounded by the leading edge portion **121**, the lower surface portion **124**, and the upper surface portion **126**.

(32) The leading edge portion **121** has a streamlined shape protruding forwards (opposite to the main wing **10**). The leading edge portion **121** is formed continuously with the upper surface portion **126**.

(33) The trailing edge portion **122** is a tip-end portion of an edge formed by a rear-end portion of the lower surface portion **124** and a rear-end portion of the upper surface portion **126**. During deployment, the trailing edge portion **122** forms a gap **G0** between the trailing edge portion **122** and the main wing **10**.

(34) The cusp portion **123** is a tip-end portion of an edge formed between a lower edge of the leading edge portion **121** and the lower surface portion **124**. The cusp portion **123** may be formed as a tip-end portion of a blade protruding toward the fixed leading edge **10a**, the blade being disposed at the lower edge of the leading edge portion **121**.

(35) The lower surface portion **124** is a concave surface formed between the cusp portion **123** and the trailing edge portion **122**. The lower surface portion **124** is a portion proximate to the fixed leading edge **10a** during retraction. The lower surface portion **124** is typically formed in a curved shape protruding forwards, corresponding to the fixed leading edge **10a**.

(36) FIG. **5** is a numerical simulation result showing an example of a flow field of the reference slat **120** shown in FIG. **4** in a landing phase.

(37) As shown in the figure, a shear layer separated from the cusp portion **123** forms a vortex-like reverse flow region on the lower surface portion **124**, and turbulence of the shear layer is generated with the formation of the reverse flow region. The generated turbulent shear layer reattaches to the lower surface portion **124** and then passes through the gap **G0** between the reference slat **120** and the main wing **10**.

(38) Noise is generated mainly by pressure fluctuations at a reattachment point **RP0** of the lower surface portion **124**, where the turbulent shear layer reattaches, and pressure fluctuations at the trailing edge portion **122**. The pressure fluctuations become the maximum at the reattachment point **RP0** and are attenuated toward the trailing edge portion **122**. The pressure fluctuations at the trailing edge portion **122** decrease with an increase in distance between the reattachment point **RP0** and the trailing edge portion **122**. In the example shown in the figure, the reattachment point **RP0** is closer to the trailing edge portion **122**. Therefore, large pressure fluctuations at the reattachment point **RP0** are not attenuated and provide large pressure fluctuations at the trailing edge portion **122**. As a result, significant noise is generated.

(39) In view of this, focusing on a strong correlation between the noise level caused by the cross-

sectional shape of the reference slat **120** and a distance **Lts0** that is the shortest distance along the lower surface from the reattachment point **RP0** of the turbulent shear layer on the lower surface portion **124** to the trailing edge portion **122**, the inventors of the present invention has achieved noise reduction by increasing the distance **Lts0** to attenuate pressure fluctuations toward the trailing edge portion **122** from the reattachment point **RP0**.

(40) Specifically, for example, the shape of the reference slat **120** is modified as follows. (1) To provide a rounded hump portion near a reattachment point of a turbulent shear layer on a slat lower surface, the rounded hump portion having a structure deformable during retraction into the main wing. (2) To extend the trailing edge portion so as to increase overlap with the main wing **10** than the reference slat **120** optimized only in terms of the aerodynamic performance.

(41) Such modifications (1) and (2) may be made independently or may be combined.

(42) [Slat According to Present Embodiment]

(43) Hereinafter, details of the slat **20** according to the present embodiment will be described with two configuration examples. Configuration Example 1 corresponds to a combination of (1) and (2) above and Configuration Example 2 corresponds to (2).

(44) It should be noted that although one slat **20** will be described as an example in the following description, the configurations may be similarly applied to all other slats **20** mounted on the fixed leading edge **10a**. In this case, the way to apply to the slat may be individually optimized depending on the slat supporting device **51** and flows at an inner end portion of the slat **20** and an outer end portion of the slat **20**.

(45) <Slat According to Configuration Example 1>

(46) FIG. **6** is a cross-sectional view of a slat **201** according to Configuration Example 1 that is the slat **20** as the leading-edge high-lift device according to the present embodiment, which is perpendicular to the wing span direction of the main wing **10** (hereinafter, also referred to as a wing span direction unless otherwise stated herein). FIG. **6** shows a state in which the slat **201** is deployed at the maximum (fully deployed) from the fixed leading edge **10a** in a landing phase.

(47) The slat **201** includes a leading edge portion **21**, a trailing edge portion **22**, a cusp portion **23**, a lower surface portion **24**, a hump portion **25**, and an upper surface portion **26**. As shown in FIG. **6**, the cross-sectional shape of the slat **201** is a shape that forms a closed space surrounded by the leading edge portion **21**, the lower surface portion **24**, and the upper surface portion **26**.

(48) The leading edge portion **21** has a streamlined shape protruding forwards (opposite to the main wing **10**). The leading edge portion **21** is formed continuously with the upper surface portion **26**.

(49) The trailing edge portion **22** is a tip-end portion of an edge formed by a rear-end portion of the lower surface portion **24** and a rear-end portion of the upper surface portion **26**. During deployment, the trailing edge portion **22** forms a gap between the trailing edge portion **22** and the main wing **10**.

(50) The cusp portion **23** is a tip-end portion of an edge formed between a lower edge of the leading edge portion **21** and the lower surface portion **24**. The cusp portion **23** may be formed at a tip-end portion of a blade protruding toward the fixed leading edge **10a**, the blade being disposed at the lower edge of the leading edge portion **21**.

(51) The lower surface portion **24** is a concave surface formed between the cusp portion **23** and the trailing edge portion **22**. The lower surface portion **24** is a portion proximate to the fixed leading edge **10a** during retraction. The lower surface portion **24** is formed in a curved shape protruding forwards, corresponding to the fixed leading edge **10a**.

(52) The hump portion **25** is locally provided on the surface of the lower surface portion **24** and has a curved shape protruding toward the main wing **10** in the cross-sectional shape perpendicular to the wing span direction.

(53) In Configuration Example 1, the trailing edge portion **22** extends toward the main wing **10**, increasing the amount of overlap with the main wing **10**, and the hump portion **25** is provided, which are differences from the reference slat **120**.

(54) As it will be described later with reference to A to C of FIG. 13, as a characteristic of the flow on the slat, the position where the turbulent shear layer in the cove (on lower surface portion) reattaches to the slat lower surface hardly change in the chord direction of the main wing, unless the flow rate of a flow passing through a gap between the slat trailing edge (TE in the figure (corresponding to the trailing edge portion 122, 22) and the main wing 10 and the position of the cusp portion are greatly changed.

(55) In the present embodiment, using this characteristic, noise reduction is achieved by setting a distance L_{ts1} between a reattachment point RP1 of the turbulent shear layer on the lower surface portion 24 and the trailing edge portion 22 to be longer than the distance L_{ts0} in the reference slat 120 so as to reduce pressure fluctuations at the trailing edge portion 22.

(56) FIG. 7 is a numerical simulation result showing an example of a flow field in the slat 201 shown in FIG. 6 in a landing phase.

(57) As shown in the figure, a shear layer separated from the cusp portion 23 forms a vortex-like reverse flow region on the lower surface portion 24, and turbulence of the shear layer is generated with the formation of the reverse flow region. The generated turbulent shear layer reattaches to the hump portion 25 and then passes through a gap G1 between the slat 201 and the main wing 10.

(58) Noise is generated mainly by pressure fluctuations at a reattachment point RP1 of the hump portion 25, where the turbulent shear layer reattaches, and pressure fluctuations at the trailing edge portion 22. In this configuration example, the reattachment point RP1 is positioned on the hump portion 25. Therefore, in addition to the trailing edge portion 22 extended longer than the trailing edge portion 122 of the reference slat 120, a distance L_{ts1} that is the shortest distance along the lower surface between the reattachment point RP1 and the trailing edge portion 22 can be set to be longer than the distance L_{ts0} (see FIG. 3) in the reference slat 120. Accordingly, large pressure fluctuations at the reattachment point RP1 can be attenuated toward the trailing edge portion 22, and pressure fluctuations at the trailing edge portion 22 can be reduced. Therefore, noise reduction can be achieved compared to the shape of the reference slat 120.

(59) (Regarding Amount of Overlap)

(60) The amount of overlap (OL amount) between the trailing edge portion 22 of the slat 201 and the main wing 10 refers to a distance (OL) between the trailing edge portion 22 and the fixed leading edge 10a, which is measured on the chord line of the slat 201 deployed at the maximum from the fixed leading edge 10a as shown in FIG. 6. As is clear from FIG. 6, the chord line of a slat 201 means a line obtained by tilting the chord line of the slat 201 in a state retracted in the main wing 10 by the deployment angle. The chord line of the slat 201 in a state retracted in the main wing 10 is along the chord line of the wing 100, not the line connecting the leading edge and the trailing edge of the slat 201.

(61) That is, assuming that C_s denotes the chord length of the slat 201 (slat chord length) and a distance (first distance) between the leading edge portion 21 and the fixed leading edge 10a, which is measured on the chord line of the slat 201 deployed at the maximum from the fixed leading edge 10a, is denoted by C_s^* , the amount of overlap OL is expressed as follows.

(62) $OL = C_s - C_s^*$

(63) Hereinafter, the first distance C_s^* will be also referred to as a reference length C_s^* .

(64) The amount of overlap OL is favorably larger to increase the distance L_{ts1} . However, if the overlap amount OL becomes too large, the noise reduction effect reaches a ceiling and the weight increases, which is unfavorable. Therefore, the amount of overlap OL is more favorably set to 35% or less of the reference length C_s^* ($OL \leq 35\% C_s^*$). That is, an optimal parameter range of the amount of overlap OL is

(65) $0\% \leq (OL / C_s^*) \leq 35\%$.

(66) The amount of extension of the trailing edge portion 22 may be uniform in the entire region in the wing span direction, at least a part of the region may be formed as a trailing edge extension portion, or a different amount of extension may be employed depending on a region. For example,

as shown in FIG. 3, a trailing edge extension portion **22e** extended to be longer than the trailing edge portion **22** may be formed in a partial region of the trailing edge portion **22** in accordance with the position of the slat supporting device **51**. In addition, the amount of extension of the trailing edge portion **22** may be optimized in the range of 35% Cs^* or less at each arbitrary position in the wing span direction.

(67) The trailing edge extension portion **22e** may be formed integrally with the trailing edge portion **22** of the slat **201**. Alternatively, the trailing edge extension portion **22e** may be additionally attached to a part of the trailing edge portion **22**. In this case, the trailing edge extension portion **22e** is configured as an adjustment member that adjusts the amount of overlap **OL** that is a distance between a tip end portion of the trailing edge extension portion **22e** and the fixed leading edge **10a**. The adjustment member may be attached to the entire region in a span direction of the trailing edge portion **22** or may be locally attached to an arbitrary region of the trailing edge portion **22**. The adjustment member is made of, for example, a metal material such as an aluminum alloy or stainless steel or a composite material such as carbon fiber reinforced plastics (CFRP) or glass fiber reinforced plastics (GFRP). The shape of the adjustment member is also not particularly limited, and may be trapezoidal, for example.

(68) Next, the hump portion **25** will be described.

(69) The hump portion **25** is provided in a reattachment region of the turbulent shear layer on the lower surface portion **24**. Forming the hump portion **25** in a curved shape protruding toward the main wing **10** allows a distance $Lts1$ that is the shortest distance along the lower surface between the reattachment point **RP1** of the turbulent shear layer and the trailing edge portion **22** to be longer than that of the reference slat **120**.

(70) The hump portion **25** is typically formed in the wing span direction. A cross-sectional shape of the hump portion **25**, which is perpendicular to the wing span direction, may be uniform in the wing span direction. Alternatively, the shape of the hump portion may be optimized in terms of noise reduction and aerodynamic and structural properties at an arbitrary position in the wing span direction depending on a change in reattachment point position in the wing span direction, pressure fluctuation distribution, and the like.

(71) For example, the thickness of the hump portion **25** is set to be smaller than in other regions, for example, in regions near inner end portions and outer end portions of the slats **20** (see FIG. 1) where the airflow structures are greatly different or regions near the slat supporting devices **51** (see FIGS. 2 and 3). In this manner, the cross-sectional shape of the hump portion **25** may differ between the region near the slat supporting devices **51** and other regions.

(72) The hump portion **25** may be formed integrally with the lower surface portion **24** (formed by curving a part of the lower surface portion **24**). Alternatively, the hump portion **25** may be an additional member different from the constituent member of the lower surface portion **24**, the additional member being added to a partial region of the lower surface portion **24**.

(73) In view of the property of retraction of the slat **201** into the fixed leading edge **10a**, the hump portion **25** is favorably configured to be deformable along the fixed leading edge **10a** during retraction as shown in FIG. 8. Accordingly, target aerodynamic performance of the aircraft wings **100** in a cruise phase can be provided. In this case, the hump portion **25** is made of a material (e.g., elastic material or shape memory alloy) or a structure having built-in various mechanism parts (e.g., link mechanisms), which is capable of bulging in a target shape during deployment and of being compressed (e.g. partially or completely flattened) during retraction.

(74) (Shape of Hump Portion)

(75) Regarding the hump portion **25**, the distance $Lts1$ between the reattachment point **RP1** of the turbulent shear layer and the trailing edge portion **22** can be set to be longer than the distance $Lts0$ in the reference slat **120** as described above. However, causing the hump portion **25** to greatly expand accelerates the flow between the hump portion **25** and the fixed leading edge **10a**. As a result, the aerodynamic performance is degraded, and increased pressure fluctuations at the

reattachment point RP1 increase noise. Thus, it is optimal to set the shape so that the reattachment point RP1 is located at a top portion 25p of the hump portion in order to minimize the hump portion 25.

(76) Moreover, as it will be described later, noise increases with an increase in pressure fluctuations at the reattachment point RP1 in a case where the curvature around the reattachment point RP1 is small or negative (it is concave as viewed from the fixed leading edge 10a). Therefore, the shape is favorably set so that the curvature around the reattachment point RP1 is a certain positive curvature (it is convex as viewed from the fixed leading edge 10a).

(77) (Thickness of Hump Portion)

(78) The top portion 25p of the hump portion 25 typically corresponds to a point of a maximum thickness δ (see FIG. 6) of the hump portion 25. The maximum thickness point δ refers to a point where the height from the lower surface portion 24 is at the maximum. The lower surface portion 24 set forth herein is a basis surface for the height. In a case where the hump portion 25 is formed integrally with the lower surface portion 24, it refers to a virtual curve that should be formed as the lower surface portion 24 if there is no hump portion 25. In a case where the hump portion 25 is formed as a member separate from the lower surface portion 24, it refers to a region of the lower surface portion 24 on which the hump portion 25 is mounted. Moreover, the maximum thickness δ of the hump portion 25 refers to a maximum value of the thickness in a direction perpendicular to such a criteria surface.

(79) (Width of Top Portion)

(80) Accounting for the fact that the curvature around a reattachment point PRI influences the noise reduction effect by the hump portion 25 and that the reattachment point RP1 moves on the surface of the hump portion 25 in accordance with a change in flight conditions (e.g., change in angle of attack) in an approach and landing phase, the top portion 25p should be considered as an area having a width on the order of the thickness of the hump portion 25. In view of this, when the maximum thickness point δ of the hump portion 25 is defined as a top C in a cross-sectional surface shape of the slat 201, which is perpendicular to the wing span direction, as shown in FIG. 9, the top portion 25p refers to a region between two points A and B on the hump portion 25 which sandwich the top C. Typically, the point A is a position spaced away from the top C toward the leading edge portion 21 along the surface of the hump portion 25 by a distance δ corresponding to the maximum thickness δ . Also, the point B is a position spaced away from the top C toward the trailing edge portion 22 along the surface of the hump portion 25 by the distance δ .

(81) (Curved Surface of Top Portion)

(82) The top portion 25p of the hump portion 25 needs to have a certain positive curvature as described above. The top portion 25p of the hump portion 25 may be formed with a curve surface having a constant curvature or may be formed with a composite curve surface whose curvature differs continuously or for each region. Moreover, the top portion 25p may include planes partially or may be formed with a curved surface made of multiple planes that makes the entire region of the top portion 25p convex (e.g., a partial polygonal surface shape inscribed in the curved surface).

(83) (Position of Top Portion)

(84) The position of the top portion 25p of the hump portion 25 is favorably determined depending on the reattachment point PRI as described above. A distance from the trailing edge portion 22 in a direction along the chord line of the slat 201 to the reattachment point PRI greatly changes depending on design conditions of the slat 201 and the maximum thickness δ of the hump portion 25. The position of the top portion 25p moves away from the trailing edge portion 22 while the maximum thickness δ increases.

(85) (Shape from Top Portion to Trailing Edge Portion)

(86) In the cross-sectional shape perpendicular to the wing span direction, a region of the hump portion 25, which is closer to the trailing edge portion 22 than the point B on the top portion 25p, as shown in FIG. 9 is favorably formed as a continuous curve in order not to increase the

turbulence at the boundary layer.

(87) It should be noted that such a curve may be formed in a curved shape that is concave with respect to the fixed leading edge **10a** in contrast to the hump portion **25** in order to obtain lift near the trailing edge portion **22** as in a case where no hump portion **25** is provided and to keep the flow velocity around the hump portion **25** low so as to prevent an unnecessary increase in pressure fluctuations at the reattachment point **RP1**.

(88) (Shape from Top Portion to Lower Surface Portion and Cusp Portion)

(89) A region from the point A of the top portion **25p** of the hump portion **25** to the lower surface portion **24** on the side of the cusp portion **23**, in the cross-sectional shape perpendicular to the wing span direction, as shown in FIG. **9**, does not necessarily need to be formed as a continuous curve because the region is in a slow reverse flow. For example, the curve of the region may have a discontinuous tangent at the point connecting to the lower surface portion **24**, as shown in FIG. **9**, may be a curve with a curvature larger than that of the top portion **25p**, or may be a bent line.

(90) Next, shape parameters of the respective parts of the hump portion **25** will be described.

(91) (Maximum Thickness of Hump Portion)

(92) In order to increase the distance **Lts1** between the reattachment point **RP1** of the shear layer and the trailing edge portion **22** so as to obtain the noise reduction effect, the maximum thickness δ of the hump portion **25** is desirably 5% or more of the reference length Cs^* (see FIG. **6**), for example. It should be noted that the maximum thickness δ of the hump portion **25** corresponds to a height of the hump portion **25** from the lower surface portion **24** at the position of the top portion **25p** of the hump portion **25**.

(93) Although increasing the maximum thickness δ of the hump portion **25** increases the distance **Lts1**, it narrows a flow channel width between the hump portion **25** and the fixed leading edge **10a**. This leads to deterioration of the aerodynamic performance of the slat **201** and an unnecessary increase in noise. Therefore, the maximum thickness δ of the hump portion **25** is favorably 15% or less of the reference length Cs^* , for example. That is, an optimal parameter range of the maximum thickness δ of the hump portion **25** is

(94) $5\% \leq (\delta / Cs^*) \leq 15\%$.

(Position of Hump Portion)

(95) Here, the position of the hump portion **25** is expressed by a distance L^* (second distance) between the fixed leading edge **10a** and the maximum thickness point (corresponding to the top portion **25p**) of the hump portion **25** in the chord direction of the slat **201** as shown in FIG. **6**.

(96) A lower limit of the distance L^* is 10% or more of the reference length Cs^* , for example, because of a limit at which the hump portion **25** with a minimum value (5% Cs^*) of the thickness δ can be provided near the trailing edge portion **22**.

(97) On the other hand, an upper limit of the distance L^* is 40% or less of the reference length Cs^* even at the maximum, accounting for a reattachment point in a typical slat design and the presence of the hump portion **25** with a maximum value (15% Cs^*) of the thickness δ .

(98) Thus, an optimal parameter range of the position of the hump portion **25** is

(99) $10\% \leq (L^* / Cs^*) \leq 40\%$.

(Average Curvature of Top Portion of Hump Portion)

(100) An average curvature of the top portion **25p** in the hump portion **25** refers to a curvature ($\kappa = f_{kids}/f_{ds}$) that is a weighted average of a local curvature κ_i in a curve AB of the hump portion **25** in the range of the top portion **25p** between the points A and B shown in FIG. **9** with a length s of the curve AB. Moreover, an average curvature radius R of the top portion **25p** is equivalent to a multiplicative inverse ($1/\kappa$) of the average curvature κ . The top portion **25p** is formed with, for example, a uniformly curved surface, combined curved surfaces, or a partially flat surface, as described above.

(101) The average curvature radius R of the top portion **25p** is favorably 50% or less of the reference length Cs^* in order to obtain the noise reduction effect by increasing the distance **Lts1**

between the reattachment point RP1 of the shear layer and the trailing edge portion 22. On the other hand, it is difficult to obtain the noise reduction effect when flight conditions change (e.g., the angle of attack changes) if the average curvature radius R is small, i.e., less than 10% of the reference length Cs*.

(102) Therefore, an optimal parameter range of the curvature radius of the top portion 25p of the hump portion 25 is

$$(103) 10\% \leq (R / Cs^*) \leq 50\%,$$

and

an optimal parameter range of the average curvature κ of the top portion 25p is

$$(104) 2 \leq Cs^* \leq 10,$$

which is double or more than double and ten times or less than ten times a multiplicative inverse of the reference length Cs*.

(105) It should be noted that in Configuration Example 1 described above, the trailing edge portion 22 extends toward the main wing 10 and the hump portion 25 is provided as differences from the reference slat 120. As a matter of course, the present invention can also be applied to a configuration example in which only the hump portion 25 is added to the reference slat 120. The addition of the hump portion 25 enables the distance along the lower surface between the reattachment point of the turbulent shear layer and the trailing edge portion to be longer than the distance Lts0 in the reference slat 120 as described above. Accordingly, large pressure fluctuations at the reattachment point can be further attenuated toward the trailing edge portion and pressure fluctuations at the trailing edge portion can be reduced. Therefore, noise level reduction can be achieved compared to the shape of the reference slat 120.

(106) <Slat According to Configuration Example 2>

(107) FIG. 10 is a cross-sectional view of a slat 202 according to Configuration Example 2 that is the slat 20 as the leading-edge high-lift device according to the present embodiment, which is perpendicular to the wing span direction of the aircraft wing 100. As in Configuration Example 1 (FIG. 6), FIG. 10 shows a state in which the slat 202 is deployed at the maximum (fully deployed) from the fixed leading edge 10a in a landing phase.

(108) The slat 202 includes a leading edge portion 21, a trailing edge portion 22, a cusp portion 23, a lower surface portion 24, and an upper surface portion 26. That is, the slat 202 according to Configuration Example 2 differs from that of Configuration Example 1 in that the slat 202 does not include the hump portion 25. Hereinafter, configurations different from those of Configuration Example 1 will be mainly described, and configurations similar to those of Configuration Example 1 will be denoted by similar reference signs and descriptions thereof will be omitted or simplified.

(109) In Configuration Example 2, the trailing edge portion 22 extends toward the main wing 10 and increases the amount of overlap with the main wing 10, which is different from the reference slat 120. Therefore, as to the trailing edge portion 22 in this configuration example, a region obtained by extending the trailing edge portion of the reference slat 120 toward the main wing 10 will be also referred to as a trailing edge extension portion 22f. A tip end portion of the trailing edge extension portion 22f corresponds to the trailing edge portion 22 extended by the trailing edge extension portion 22f.

(110) As to the amount of extension of the trailing edge portion 22 by the trailing edge extension portion 22f, as shown in FIG. 10, assuming that a distance (first distance) between the leading edge portion 21 and the fixed leading edge 10a, which is measured on the chord line of the slat 202 during maximum deployment from the fixed leading edge 10a, is a reference distance Cs*, the amount of overlap OL that is a distance between the tip end portion of the trailing edge extension portion 22f and the fixed leading edge 10a, which is measured on the chord line of the slat 202 during maximum deployment from the fixed leading edge 10a, is set to 10% or more and 35% or less of the reference distance Cs* as it will be described later.

(111) FIG. 11 is a numerical simulation result showing an example of a flow field in the slat 202

shown in FIG. 10 in a landing phase.

(112) As shown in the figure, a shear layer separated from the cusp portion 23 forms a vortex-like reverse flow region on the lower surface portion 24, and turbulence of the shear layer is generated with the formation of the reverse flow region. The generated turbulent shear layer reattaches to the lower surface portion 24 and then passes through a gap G2 between the slat 202 and the main wing 10.

(113) Noise is generated mainly by pressure fluctuations at a reattachment point RP2 on the lower surface portion 24 where the turbulent shear layer reattaches and pressure fluctuations at the trailing edge portion 22. In this configuration example, the trailing edge portion 22 is extended longer than the trailing edge portion 122 of the reference slat 120 by the trailing edge extension portion 22f. Therefore, a distance Lts2 that is the shortest distance along the lower surface between the reattachment point RP2 and the trailing edge portion 22 can be set to be longer than the distance Lts0 (see FIG. 4) in the reference slat 120. Accordingly, large pressure fluctuations at the reattachment point RP2 can be attenuated toward the trailing edge portion 22 and pressure fluctuations at the trailing edge portion 22 can be reduced. Therefore, noise level reduction can be achieved compared to the shape of the reference slat 120.

(114) (Regarding Amount of Overlap)

(115) An amount of overlap OL between the trailing edge portion 22 of the slat 202 and the main wing 10 refers to a distance between the trailing edge portion 22 and the fixed leading edge 10a, which is measured on the chord line of the slat 201 deployed at the maximum from the fixed leading edge 10a, as in Configuration Example 1 (see FIG. 10).

(116) The amount of overlap OL is favorably at least 10% or more and ($OL \geq 10\% Cs^*$) of the reference length Cs^* in order to obtain the noise reduction effect because the distance Lts2 for compensating for the absence of the hump portion 25 is needed. However, if the overlap amount OL becomes too large for increasing the distance Lts2, the noise reduction effect reaches a ceiling and the weight increases, which is unfavorable. The amount of overlap OL is favorably set to 35% or less ($OL \leq 35\% Cs^*$) of the reference length Cs^* . Accordingly, noise can be effectively reduced while ensuring the aerodynamic performance, which is the objective of the slat 202.

(117) On the other hand, the amount of overlap OL tends to be 10% Cs^* or more on an inboard side of the aircraft wing 100 relative to the engine 40. However, effective noise reduction by extending the trailing edge portion 22 cannot be exhibited because a relative ratio ($Cs^*/Cstow$) of the reference length Cs^* to the chord length Cstow (see FIG. 8) of the aircraft wing 100 on a cruise mode on which the slats 20 and the flaps 30 are retracted is 10% or less, resulting in the slats 20 as a whole too close to the main wing 10. Moreover, if the relative ratio ($Cs^*/Cstow$) is 20% or more, the slats 20 are structurally too large relative to the main wing 10.

(118) Therefore, an optimal parameter range of the amount of overlap OL in Configuration Example 2 is

(119) $10\% \leq (OL / Cs^*) \leq 35\%$ and $10\% \leq (Cs^* / Cstow) \leq 20\%$.

(120) Also, in this configuration example, the amount of extension of the trailing edge portion 22 may be uniform in the wing span direction or at least a partial region may be formed as the trailing edge extension portion 22f. The trailing edge extension portion 22f may be formed as a separate member added to a trailing edge portion of the existing slat. For example, FIG. 12 schematically shows, as another configuration example of the leading-edge high-lift device (slat 20) according to the present embodiment, a slat 203 obtained by adding to the trailing edge portion 122 of the reference slat 120 shown in FIG. 4 an adjustment member 150 capable of adjusting its amount of extension. In FIG. 12, the adjustment member 150 forms the trailing edge portion 22 by extending the trailing edge portion 122 of the reference slat 120 by an arbitrary amount of adjustment W. The adjustment member 150 forms the trailing edge extension portion 22f on the trailing edge portion 122 by being arranged to overlap on a part of the lower surface portion 124 or the upper surface portion 126 as shown in the figure. The adjustment member 150 is made of, for example, a metal

material such as an aluminum alloy or stainless steel or a composite material such as carbon fiber reinforced plastics (CFRP) or glass fiber reinforced plastics (GFRP). The shape of the extended region of the trailing edge portion is also not particularly limited. In a case where the extended region of the trailing edge portion is arranged in a partial region, the extended region of the trailing edge portion has a trapezoidal shape, for example, as in the trailing edge extension portion **22e** shown in FIG. 3.

(121) In the slat **203** shown in FIG. 12, as in Configuration Example 2, an optimal parameter range of the amount of overlap OL, which is a distance between the tip end portion of the trailing edge extension portion **22f** and the fixed leading edge **10a** measured on the chord line of the slat **203** at the maximum deployment from the fixed leading edge **10a**, is:

(122) $10\% \leq (OL / Cs^*) \leq 35\%$ and $10\% \leq (Cs^* / C_{stow}) \leq 20\%$.

<Characteristic Assessment>

(123) FIG. 13 is numerical simulation results showing pressure fluctuations of a slat trailing edge portion, which cause noise generation, in which A shows an example of the reference slat **120**, B shows an example of the slat **201** according to Configuration Example 1, and C shows an example of the slat **202** according to Configuration Example 2.

(124) Here, the distance (Lts0) between the reattachment point RP0 of the shear layer on the reference slat **120** and the trailing edge portion **122** was 13% Cs^* .

(125) Moreover, the distance (Lts1) between the reattachment point RP1 of the shear layer and the trailing edge portion **22** was 32% Cs^* when the maximum thickness δ of the hump portion **25** was set to 7% Cs^* and a trailing edge extension amount was set to 11% Cs^* in the slat **201** according to Configuration Example 1.

(126) In addition, the distance (Lts2) between the reattachment point RP2 of the shear layer and the trailing edge portion **22** was 33% Cs^* when a trailing edge extension amount was set to 20% Cs^* in the slat **202** according to Configuration Example 2.

(127) As shown in A to C of FIG. 13, as a characteristic of the flow on the slat, the position (reattachment point RP0, RP1, RP2) where the turbulent shear layer in the cove (on lower surface portion) reattaches to the slat lower surface hardly changes in the chord direction of the aircraft wing unless the flow rate of a flow passing through a gap between the slat trailing edge (TE in the figure (corresponding to the trailing edge portion **122**, **22**) and the main wing **10** and the position of the cusp portion greatly change.

(128) In the present embodiment, using this characteristic, pressure fluctuations of the slat trailing edge TE (trailing edge portion **22**), which cause noise, are reduced.

(129) That is, in the present invention, as shown in FIGS. 7 and 11, the distance Lts1 and Lts2 from the reattachment point RP1 and RP2 to the trailing edge portion **22** is increased by providing the hump portion **25** to the slat lower surface or extending the trailing edge portion **22**, pressure fluctuations toward the trailing edge portion **22** from the reattachment point RP1 and RP2 are further attenuated, and pressure fluctuations at the trailing edge portion **22** are reduced.

(130) As a result, as shown in FIGS. 13 B and C, it can be seen that the pressure fluctuations at the slat trailing edge TE decrease in comparison with the pressure fluctuations of the reference slat **120** (FIG. 13A).

(131) Next, FIG. 14 shows a numerical simulation result of overall sound pressure level (OASPL) directivity when slat noise was assessed on the scale of a regional jet airliner. Seven types of slats, as follows, were used for the analysis. Slat 1: a reference slat (OL amount=−7% Cs^*) Slat 2: a slat with a trailing edge extension amount of 7% Cs^* (OL amount=1% Cs^*) Slat 3: a slat with a trailing edge extension amount of 11% Cs^* (OL amount=5% Cs^*) Slat 4: a slat with a trailing edge extension amount of 20% Cs^* (OL amount=14% Cs^*) Slat 5: a slat with a trailing edge extension amount of 7% Cs^* and a hump portion (average curvature $\kappa=4.1/Cs^*$) Slat 6: a slat with a trailing edge extension amount of 11% Cs^* and a hump portion (average curvature $\kappa=4.3/Cs^*$) Slat 7: a slat with a trailing edge extension amount of 20% Cs^* and a hump portion (average curvature

$\kappa=4.9/Cs^*$)

(132) As shown in FIG. 14, as to the slats 2 to 7, significant noise reduction is found in all directions in comparison with the slat 1. In particular, a large noise reduction effect is obtained in the ground direction (at 270 degrees): 4 dB for the slat 2, 3 dB for the slat 3, 9 dB for the slat 4, 8 dB for the slat 5, 10 dB for the slat 6, and 12 dB for the slat 7. That is, the slats 2 to 4, which only extend the trailing edge, reduce noise by 4 dB to 9 dB, and the slats 5 to 7, which add the hump portion (average curvature $4/Cs^*$ or more), further reduce noise by about 4 dB.

(133) It should be noted that in the assessment example shown in FIG. 14, the OL amount of the reference slat refer to the slat 1 was set to $-7\% Cs^*$. However, the noise reduction effect by 4 dB or more cannot be expected with the OL amount of the slat 2 or 3 because the OL amount of the existing slat is often around $0\% Cs^*$. Thus, in order to obtain a significant noise reduction effect for the reference slat with an OL amount of around $0\% Cs^*$, a trailing edge extension amount for the reference slat is favorably $10\% Cs^*$ or more in a case where the hump portion is not provided.

(134) FIG. 15 shows a numerical simulation result of lift characteristics for the slats 1, 3, 4, and 6. In FIG. 15, the horizontal axis indicates an angle of attack and the vertical axis indicates a lift coefficient. As shown in the figure, it has been confirmed that all the slats 1, 3, 4, and 6 provide substantially the same maximum lift as the reference-shape slat 1 and can maintain the functions of the slats as the high-lift devices.

(135) Next, FIG. 16 is a graph showing the noise reduction effects of the slats 2 to 7, comparing the overall sound pressure level at 270 degrees with the slat 1 (reference slat) shown in FIG. 14. In the figure, the horizontal axis indicates a relative ratio (Lts/Cs^*) of the distance Lts between the reattachment point and the trailing edge portion to the reference length Cs^* (see FIG. 6) and the vertical axis indicates a difference from the sound pressure level of the slat 1 at 270 degrees.

(136) As shown in FIG. 16, the noise reduction effect tends to be enhanced with a slat shape at a higher relative ratio of the distance Lts to the reference length Cs^* .

(137) Moreover, it has been found that as to the slat shape according to Configuration Example 1 with the hump portion 25, the level of the noise reduction effect varies depending on the average curvature κ of the top portion 25p of the hump portion 25, and typically, a more significant noise reduction effect is obtained with an increase in the average curvature κ .

(138) In addition, as a result of checking the noise reduction effect by arbitrarily adjusting the average curvature κ in the slats 5 to 7, as shown in FIG. 16, it has been found that setting the average curvature κ to be $2.0/Cs^*$ or more is more effective for large noise reduction than the slats 2 to 4 with the same trailing edge extension amount. In particular, the average curvature is favorably set to provide a noise reduction effect equivalent or nearly equivalent to an estimated noise curve shown as the broken line in the figure.

Another Embodiment

(139) Next, another embodiment of the present invention will be described with reference to FIG. 17. In the present embodiment, an application example to a Krueger flap 320 as the high-lift device will be described.

(140) FIG. 17 is a cross-sectional view of the Krueger flap 320 as the leading-edge high-lift device according to the present embodiment, which is perpendicular to a wing span direction of an aircraft wing 300. FIG. 17 shows a state in which the Krueger flap 320 is deployed at the maximum (fully deployed) from a fixed leading edge 310a.

(141) The Krueger flap 320 includes a leading edge portion 321, a trailing edge portion 322, a cusp portion 323, a lower surface portion 324, and a hump portion 325, as in the slat 201. The Krueger flap 320 is made of, for example, a metal material such as an aluminum alloy or stainless steel or a composite material such as carbon fiber reinforced plastics (CFRP) or glass fiber reinforced plastics (GFRP).

(142) The trailing edge portion 322 forms a gap between the trailing edge portion 322 and a main wing 310 during deployment.

- (143) The cusp portion **323** is formed at a lower edge of the leading edge portion **321**.
- (144) The lower surface portion **324** is a concave surface formed between the cusp portion **323** and the trailing edge portion **322**.
- (145) The hump portion **325** is locally provided on a surface of the lower surface portion **324** and has a curved shape protruding toward the main wing **310** in the cross-sectional shape perpendicular to the wing span direction.
- (146) The hump portion **325** is provided in a region of the lower surface portion **324** where a turbulent shear layer reattaches. Forming the hump portion **325** in the curved shape protruding toward the main wing **310** enables the distance L_{ts} that is a distance along the lower surface between the reattachment point (corresponding to RP1 in FIG. 6) of the turbulent shear layer and a trailing edge **322**. Accordingly, pressure fluctuations at the trailing edge portion **322** can be reduced and noise reduction can be achieved.
- (147) Details of the hump portion **325** are similar to the hump portion **25** on the slat **201**, so a description thereof will be omitted. It should be noted that the hump portion **325** does not need to be configured to be deformable during retraction into the main wing **310** because the Krueger flap **320** is configured to be stowed in a storage portion **311** mounted on the lower surface of the main wing **310**.
- (148) The trailing edge portion **322** can be configured to be similar to the trailing edge portion **22** in the above-mentioned slat **201** or **202**. Also in this case, the trailing edge portion **322** may be extended toward the main wing **10** so as to increase the amount of overlap with the main wing **310**. Accordingly, the distance L_{ts} between the reattachment point of the turbulent shear layer on the lower surface portion **324** and the trailing edge portion **322** can be increased. Therefore, pressure fluctuations at the trailing edge portion **322** can be reduced and noise reduction can be achieved.
- (149) Furthermore, the adjustment member (see FIG. 12) that adjusts the amount of overlap with the fixed leading edge **10a** may also be applied in the present embodiment. Accordingly, adding the adjustment member to the existing Krueger flaps can provide mechanisms and effects similar to those described above.

REFERENCE SIGNS LIST

(150) **10, 310** main wing **20, 201, 202, 203** slat (leading-edge high-lift device) **21, 321** leading edge portion **22, 322** trailing edge portion **22e, 22f** trailing edge extension portion **23, 323** cusp portion **24, 324** lower surface portion **25, 325** hump portion **150** adjustment member **320** Krueger flap (leading-edge high-lift device)

Claims

1. A leading-edge high-lift device that is deployable and retractable from/into a fixed leading edge of a main wing of an aircraft, comprising: a leading edge portion; a trailing edge portion that forms a gap between a trailing edge portion and the main wing during deployment; a cusp portion formed at a lower edge of the leading edge portion; a lower surface portion formed between the cusp portion and the trailing edge portion; and a curved hump portion that is locally provided on a surface of the lower surface portion and protrudes toward the main wing in a cross-sectional shape perpendicular to a wing span direction of the main wing, wherein, assuming that in a cross-sectional shape perpendicular to the wing span direction of the main wing, a distance between the leading edge portion and the fixed leading edge, which is measured on a chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is a first distance and a distance between a top portion of the hump portion and the fixed leading edge, which is measured on the chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is a second distance, the second distance is 10% or more and 40% or less of the first distance.
2. A leading-edge high-lift device that is deployable and retractable from/into a fixed leading edge

of a main wing of an aircraft, comprising: a leading edge portion; a trailing edge portion that forms a gap between a trailing edge portion and the main wing during deployment; a cusp portion formed at a lower edge of the leading edge portion; a lower surface portion formed between the cusp portion and the trailing edge portion; and a curved hump portion that is locally provided on a surface of the lower surface portion and protrudes toward the main wing in a cross-sectional shape perpendicular to a wing span direction of the main wing, wherein, assuming that in a cross-sectional shape perpendicular to the wing span direction of the main wing, a distance between the leading edge portion and the fixed leading edge, which is measured on a chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is a first distance, a maximum thickness of the hump portion from the surface of the lower surface portion is 5% or more and 15% or less of the first distance.

3. A leading-edge high-lift device that is deployable and retractable from/into a fixed leading edge of a main wing of an aircraft, comprising: a leading edge portion; a trailing edge portion that forms a gap between a trailing edge portion and the main wing during deployment; a cusp portion formed at a lower edge of the leading edge portion; a lower surface portion formed between the cusp portion and the trailing edge portion; and a curved hump portion that is locally provided on a surface of the lower surface portion and protrudes toward the main wing in a cross-sectional shape perpendicular to a wing span direction of the main wing, wherein, assuming that in a cross-sectional shape perpendicular to the wing span direction of the main wing, a distance between the leading edge portion and the fixed leading edge, which is measured on a chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is a first distance, an average curvature of a top portion of the hump portion is double or more than double and ten times or less than ten times a multiplicative inverse of the first distance.

4. A leading-edge high-lift device that is deployable and retractable from/into a fixed leading edge of a main wing of an aircraft, comprising: a leading edge portion; a trailing edge portion that forms a gap between a trailing edge portion and the main wing during deployment; a cusp portion formed at a lower edge of the leading edge portion; a lower surface portion formed between the cusp portion and the trailing edge portion; and a curved hump portion that is locally provided on a surface of the lower surface portion and protrudes toward the main wing in a cross-sectional shape perpendicular to a wing span direction of the main wing, wherein, assuming that in a cross-sectional shape perpendicular to the wing span direction of the main wing, a distance between the leading edge portion and the fixed leading edge, which is measured on a chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is a first distance, an amount of overlap that is a distance between the trailing edge portion and the fixed leading edge, which is measured on the chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is 35% or less of the first distance.

5. The leading-edge high-lift device according to claim 4, further comprising an adjustment member that is attached to at least a part of the trailing edge portion and adjusts the amount of overlap.

6. The leading-edge high-lift device according to claim 1, wherein the hump portion is configured to expand during deployment from the fixed leading edge and be deformable along the fixed leading edge during retraction into the fixed leading edge.

7. A leading-edge high-lift device that is deployable and retractable from/into a fixed leading edge of a main wing of an aircraft, comprising: a leading edge portion; a trailing edge portion that forms a gap between a trailing edge portion and the main wing during deployment; a cusp portion formed at a lower edge of the leading edge portion; a lower surface portion formed between the cusp portion and the trailing edge portion; and a trailing edge extension portion provided on at least a part of the trailing edge portion, wherein assuming that in a cross-sectional shape perpendicular to the wing span direction of the main wing, a distance between the leading edge portion and the fixed leading edge, which is measured on a chord line of the leading-edge high-lift device during

maximum deployment from the fixed leading edge, is a first distance, an amount of overlap that is a distance between a tip end portion of the trailing edge extension portion and the fixed leading edge, which is measured on the chord line of the leading-edge high-lift device during maximum deployment from the fixed leading edge, is 10% or more and 35% or less of the first distance, and the first distance is 10% or more and 20% or less of a chord length of an aircraft wing during retraction of the leading-edge high-lift device into the fixed leading edge.

8. The leading-edge high-lift device according to claim 7, wherein the trailing edge extension portion is an adjustment member that is attached to the trailing edge portion and adjusts the amount of overlap.

9. A wing comprising the leading-edge high-lift device according to claim 1.

10. An aircraft comprising the leading-edge high-lift device according to claim 1.

11. A wing comprising the leading-edge high-lift device according to claim 7.

12. An aircraft comprising the leading-edge high-lift device according to claim 7.
